STA 3402 – Machine Learning & Neural Networks

Lecture 3 – Dimension Reduction for Categorical and Mixed Data: MCA and FAMD
(plus an introduction to Cluster Analysis)

Full study notes from the lecture R code, Tutorial 1 (MCA), the Self-Learning Activity and the two data sets (simple English).
All numbers and plots in these notes were produced by actually running the R code on your data files (FactoMineR 2.9, factoextra 1.0.7).

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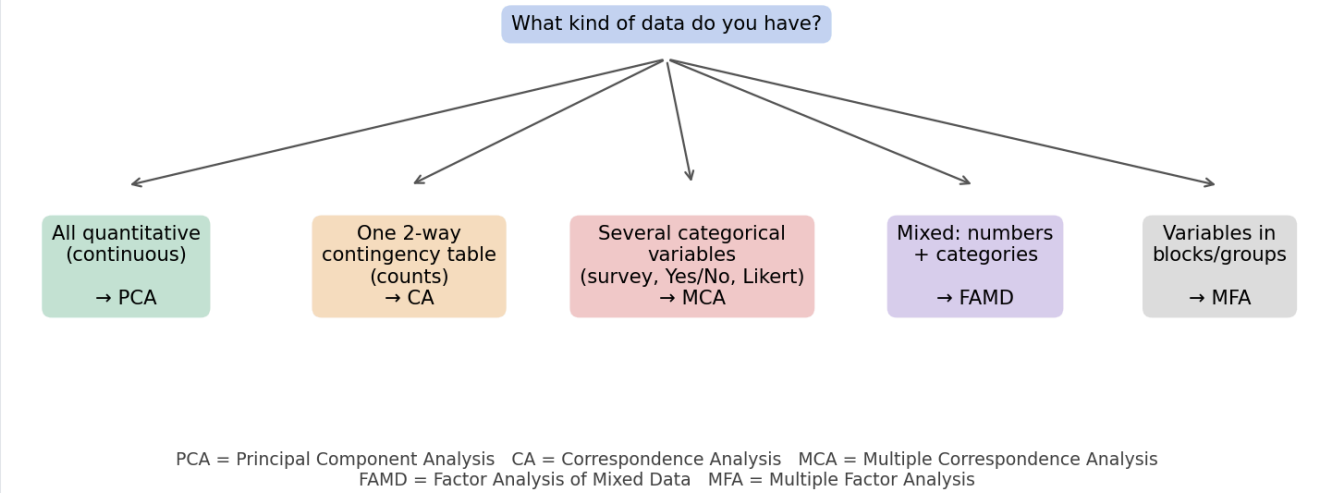
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1. Why PCA is not enough for categorical data

Last week we learnt **PCA**. PCA works only for **quantitative (numeric)** variables and uses ordinary (Euclidean) distance. At the end of the last lecture the question was: **what if all (or some) of our variables are categorical?**

- **Nominal** categories (e.g. Study_Mode = Solo / Group / Mixed) have **no order** and **no numeric distance**. "Group = 1, Mixed = 2, Solo = 3" would be meaningless numbers.
- Even for **ordered** categories like Low / Medium / High, replacing them with 1, 2, 3 **forces equal gaps** (Low→Medium = Medium→High) and treats them as exact measurements. This **changes the meaning of distance**, which we did not measure.
- So we need methods designed for categories: **MCA** (all categorical) and **FAMD** (mixed numeric + categorical).

2. Choosing a method



Method	Use when…	R function (FactoMineR)
PCA	All variables are quantitative (continuous)	`PCA()` or `prcomp()`
CA – Correspondence Analysis	One contingency table of two categorical variables (counts), e.g. district × transport mode	`CA()`
MCA – Multiple Correspondence Analysis	Several categorical variables (surveys, Yes/No, Likert)	`MCA()`
FAMD – Factor Analysis of Mixed Data	A mix of quantitative and categorical variables	`FAMD()`
MFA – Multiple Factor Analysis	Variables are organised in **groups/blocks** (e.g. genetics, clinical, imaging, lifestyle)	`MFA()`

Packages: `install.packages(c("FactoMineR", "factoextra"))`. **FactoMineR** does the analysis; **factoextra** makes nice ggplot2 graphs (functions starting with `fviz_`).

3. MCA – the idea

Conceptually, MCA does three things:

1. It creates an **indicator (disjunctive) table**: one 0/1 column for every category.
2. It compares **category profiles** (which students chose which categories).
3. It finds a small number of **dimensions** that summarise how students **depart from the average profile**.

3.1 The indicator (disjunctive) table – example

Five students from our data, with two variables:

Student	Study_Mode	Attendance
S001	Solo	Regular
S003	Group	Regular
S007	Solo	Occasional
S009	Mixed	Occasional
S021	Solo	Rare

	Group	Mixed	Solo	Occ.	Rare	Reg.
S001	0	0	1	0	0	1
S003	1	0	0	0	0	1
S007	0	0	1	1	0	0
S009	0	1	0	1	0	0
S021	0	0	1	0	1	0

Each row has exactly one "1" per variable. In R you can see this with `FactoMineR::tab.disjonctif(data)`. MCA is basically a **Correspondence Analysis of this 0/1 table**.

- A **student** is represented by the set of categories they have.
- Categories chosen by **similar students** end up in **similar directions** in the map.
- **Rare categories** are "unusual", so they lie **far from the origin** and can have big influence.

3.2 Words used in MCA

Word	Meaning
Inertia	The "variation" in categorical data (like variance in PCA). Total inertia = $(K/J) - 1$, where K = number of categories and J = number of variables.
Eigenvalue	Inertia explained by a dimension. In MCA each eigenvalue is ≤ 1. Number of dimensions = $K - J$.
Coordinate	Position of a category/student on a dimension.
Contribution (contrib)	How much (%) a category helps to **build** a dimension. All contributions on one dimension add to 100%.
cos2 (squared cosine)	How **well** a category/student is **represented** on a dimension or plane (0 to 1). Closer to 1 = better.
eta2 (η^2)	For a whole variable: squared correlation ratio between the variable and a dimension (0 to 1).
v.test	A z-like test value. $\|v.test\| > 2$ (about 1.96) means the category is significantly different from 0 on that dimension.
Supplementary variable	A variable that is **projected** onto the map afterwards but **does not** help to build the axes.

Interpretation warning (from the tutorial)

- Categories **near the origin** tend to be non-discriminating (common, "average").
- Categories **far from the origin** may help define dimensions.
- Categories in a **similar direction** can indicate similar response profiles.
- A **short plotted distance does not automatically prove a strong relationship**. Use contribution, cos2, frequencies and subject knowledge together.

4. Tutorial 1: MCA step by step

Data: `student_learning_profiles_categorical.csv` – 180 students, 12 columns: `Student_ID`, `Programme`, `Study_Mode`, `Resource_Preference`, `Attendance`, `Assignment_Timing`, `Coding_Confidence`, `Statistics_Confidence`, `AI_Assistant_Use`, `Preferred_Assessment`, `Primary_Device`, `Performance_Band`. All are categorical.

- **Active variables** (build the axes): the 9 learning behaviour / preference variables.
- **Supplementary variables** (only projected): `Programme` and `Performance_Band`.

4.1 Read the data

```
dat <- read.csv("student_learning_profiles_categorical.csv",
                stringsAsFactors = TRUE)
str(dat)
head(dat)
```

- `stringsAsFactors = TRUE` turns every text column into a **factor** (categorical variable). MCA needs factors.
- `str(dat)`: 180 obs. of 12 variables, all factors (e.g. Programme: 4 levels; Attendance: 3 levels).
- Save the CSV in your working directory, or give the full path / use `setwd()`.

> **A side line in the lecture code**
> `read.csv("data2.txt", sep=";")` shows how to read a file whose values are separated by **semicolons** instead of commas. `sep` tells R the separator. (`read.csv2()` also uses ";" by default.)

```
# remove the identifier from the analysis
rownames(dat) <- dat$Student_ID
dat$Student_ID <- NULL

# check variable types
sapply(dat, class)
# if any categorical column is still character, convert it
dat[] <- lapply(dat, as.factor)
```

- The ID is **not** a variable – it only names each student. We move it into the **row names** (so plots can still label students) and delete the column. Otherwise MCA would treat 180 IDs as 180 categories!
- `sapply(dat, class)` shows the class of every column → all "factor".
- `dat[] <- lapply(dat, as.factor)` converts **every** column to factor. The `[]` keeps it as a data frame (without it you would get a list).

```
library(FactoMineR)
library(factoextra)
?MCA       # help page
?mca       # does NOT work – R is case-sensitive; the function is MCA()
```

4.2 Explore category frequencies first

```
lapply(dat, table)
round(prop.table(table(dat$Attendance)) * 100, 1)
```

Dimension reduction does **not** remove the need for basic exploratory analysis. Results from your data:

Variable	Categories (counts, n = 180)
Programme *(sup.)*	Biological Science 31, Data Science 59, Physical Science 42, Statistics 48
Study_Mode	Group 47, Mixed 71, Solo 62
Resource_Preference	Interactive Tools 55, Lecture Notes 59, Videos 66
Attendance	Occasional 51, **Rare 12**, Regular 117
Assignment_Timing	Early 51, Last Minute 33, On Time 96
Coding_Confidence	High 74, Low 31, Medium 75
Statistics_Confidence	High 80, Low 32, Medium 68
AI_Assistant_Use	Frequent 96, **Never 16**, Occasional 68
Preferred_Assessment	Exam 57, Project 86, Quiz 37
Primary_Device	Laptop 125, Mobile 40, **Shared Computer 15**
Performance_Band *(sup.)*	High 83, Low 27, Medium 70

Attendance in %: Occasional 28.3%, Rare 6.7%, Regular 65.0%.

Checkpoint 1 – answers

1. **Three categories:** all active variables (Study_Mode, Resource_Preference, Attendance, Assignment_Timing, Coding_Confidence, Statistics_Confidence, AI_Assistant_Use, Preferred_Assessment, Primary_Device) and Performance_Band. Only Programme has four.
2. **Rare categories:** Attendance = Rare (12, 6.7%), Shared Computer (15, 8.3%), AI = Never (16, 8.9%). Rare categories get large coordinates and can dominate an axis, so interpret them carefully.
3. **Similar patterns (guess before MCA):** Rare attendance, Last Minute submission, Low confidence and Mobile device may go together; Regular attendance, Early submission, High confidence and Laptop may go together.
4. **Low/Medium/High as 1/2/3:** this assumes equal, measured gaps and turns an ordinal label into a number. Euclidean distance would then depend on an arbitrary coding, which changes the meaning of distance. MCA treats each category separately and needs no such assumption.

4.3 Run the MCA

```
sup_idx <- which(names(dat) %in% c("Programme", "Performance_Band"))
sup_idx          # 1 11  (column numbers)

res_mca <- MCA(
  dat,
  quali.sup = sup_idx,   # supplementary categorical variables
  ncp = 5,               # keep 5 dimensions in the results
  graph = FALSE          # don't draw plots now (TRUE draws default plots)
)
res_mca
```

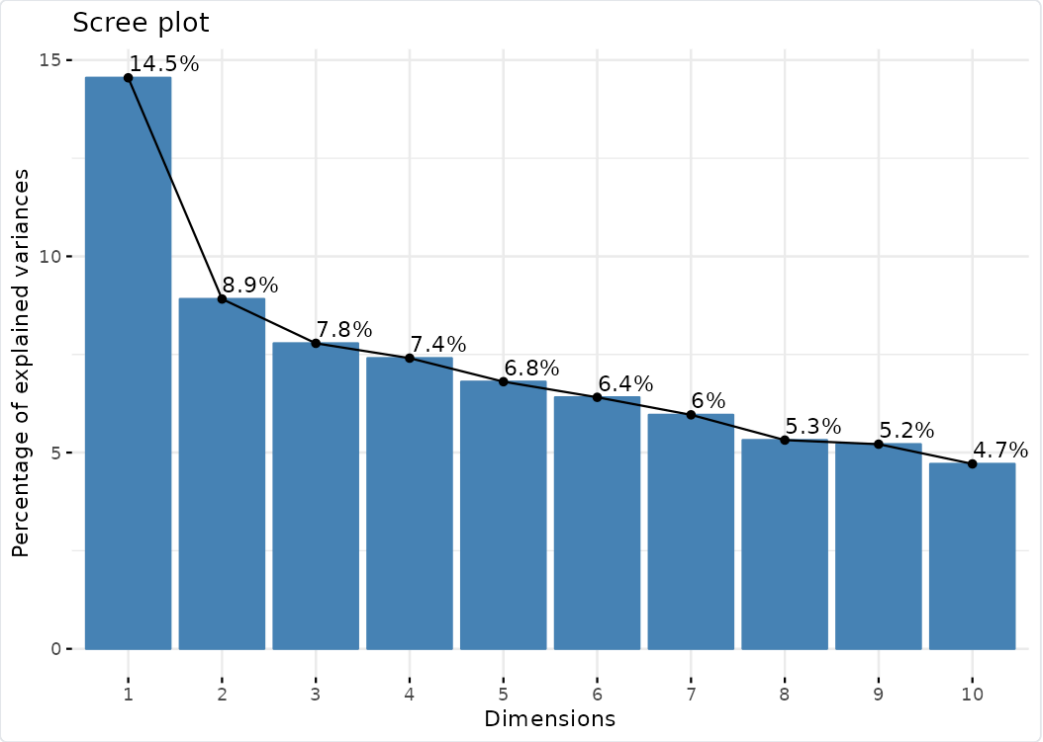
- `names(dat) %in% c(...)` gives TRUE/FALSE for each column name; `which()` returns the positions of the TRUEs → 1 and 11.
- `quali.sup` = supplementary **qualitative** variables. They are projected onto the final map but **do not influence** where the axes are placed. This lets us check whether an external grouping (e.g. performance) lines up with the structure found from behaviour.
- Printing `res_mca` lists everything inside: `$eig`, `$var$coord`, `$var$cos2`, `$var$contrib`, `$var$v.test`, `$var$eta2`, `$ind$coord`, `$ind$cos2`, `$ind$contrib`, `$quali.sup$coord`, `$quali.sup$cos2`, `$quali.sup$v.test`…
- With `graph = TRUE` (also in the lecture code), MCA draws its default factor maps immediately.

4.4 Eigenvalues and dimensionality

```
res_mca$eig
fviz_eig(res_mca, addlabels = TRUE)
```

Dimension	Eigenvalue	% of inertia	Cumulative %
Dim 1	0.2910	14.55	14.55
Dim 2	0.1783	8.91	23.46
Dim 3	0.1557	7.78	31.25
Dim 4	0.1481	7.41	38.65
Dim 5	0.1361	6.81	45.46
…	…	…	…
Dim 18	0.0433	2.16	100.00

There are 9 active variables (J) with 27 categories (K), so there are K − J = **18 dimensions**. Total inertia = 27/9 − 1 = 2, and 0.2910/2 = 14.55%.



Checkpoint 2 – answers

1. **Dim 1** explains the largest share: 14.55% of inertia.
2. There is a clear drop after Dim 1 (14.5% → 8.9%), then Dims 2–7 fall slowly (8.9, 7.8, 7.4, 6.8, 6.4, 6.0%). So there is **no very sharp elbow after Dim 2**; the main "elbow" is after Dim 1 or 2. (A common MCA rule keeps dimensions with eigenvalue above the average 1/J = 1/9 = 0.111 → that would keep 7.)
3. **No.** Two dimensions (23.5%) are good for a readable map and interpretation, but they are **not "the data"**. For clustering or prediction you may keep more (e.g. 5), because much information is in later dimensions.

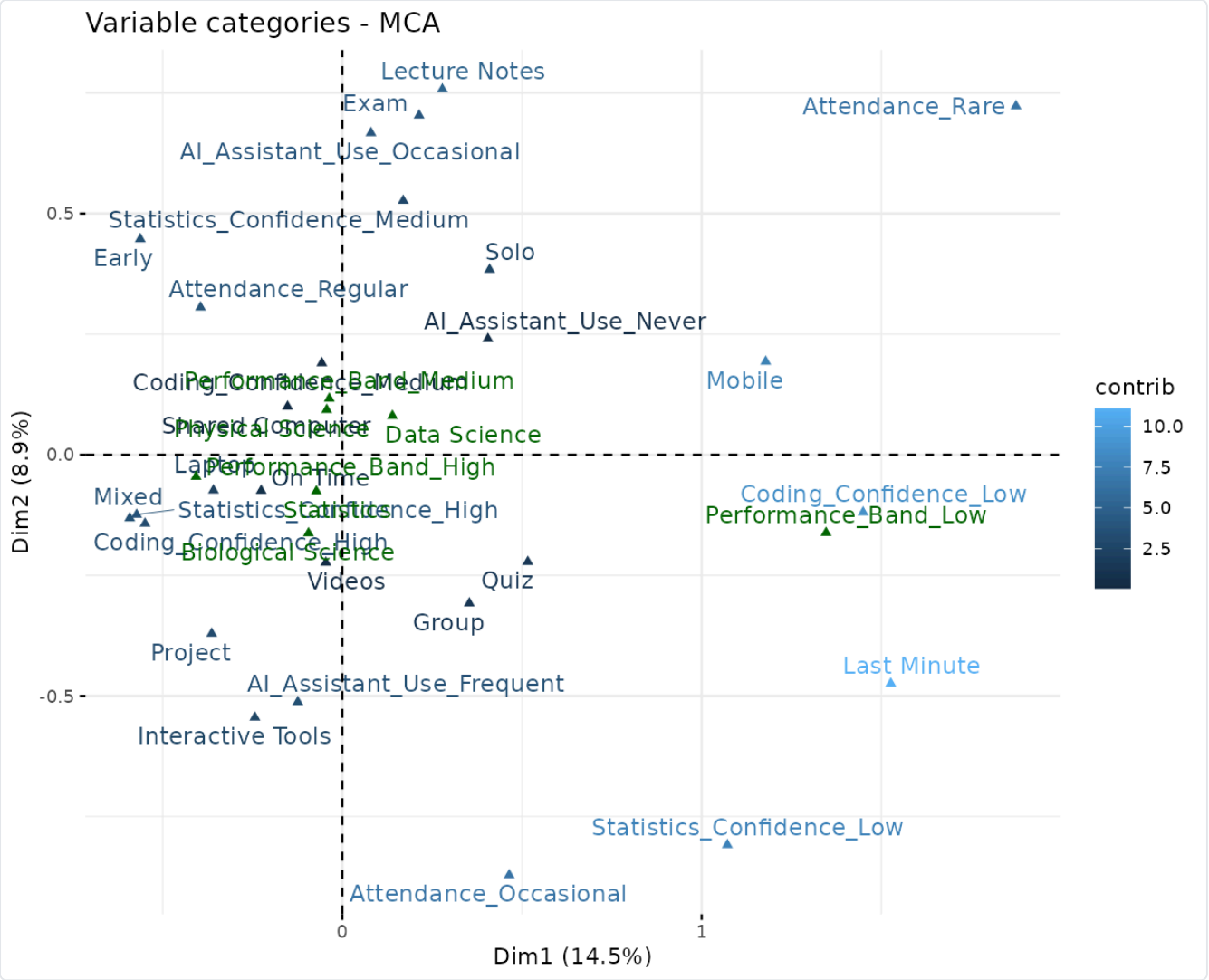
Note on percentage inertia

MCA percentages are usually **low** compared with PCA, because the 0/1 coding creates many artificial dimensions. Do **not** apply an automatic "keep dimensions until 80%" rule. Think about the coding and the goal. (Corrected versions such as Benzécri's adjustment give higher, more realistic percentages.)

4.5 Plot the categories

```
fviz_mca_var(res_mca, choice = "var.cat", repel = TRUE)
fviz_mca_var(res_mca, choice = "var.cat", col.var = "contrib", repel = TRUE)
```

- `choice = "var.cat"` plots the **categories**. `repel = TRUE` avoids overlapping labels. `col.var = "contrib"` colours each category by its contribution (lighter blue = larger). Supplementary categories are shown in green.



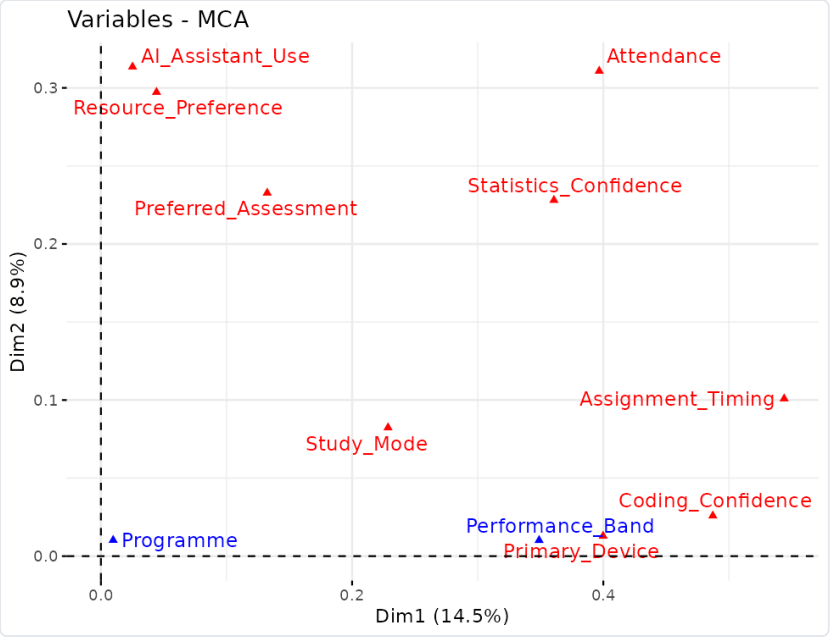
MCA category map coloured by contribution (your data)

Reading the map:

- **Right side of Dim 1:** Last Minute, Coding_Confidence_Low, Mobile, Attendance_Rare, Statistics_Confidence_Low – and the supplementary Performance_Band_Low.
- **Left side of Dim 1:** Early, Attendance_Regular, Coding/Statistics_Confidence_High, Laptop, Mixed study mode, Project – near Performance_Band_High.
- **Top of Dim 2:** Lecture Notes, Exam, AI_Assistant_Use_Occasional, Statistics_Confidence_Medium, Solo.
- **Bottom of Dim 2:** Attendance_Occasional, Statistics_Confidence_Low, Interactive Tools, AI_Assistant_Use_Frequent, Project.

Another useful view, the **variables** map (η^2 of each variable on Dims 1 and 2):

```
fviz_mca_var(res_mca, choice = "var", repel = TRUE)
```

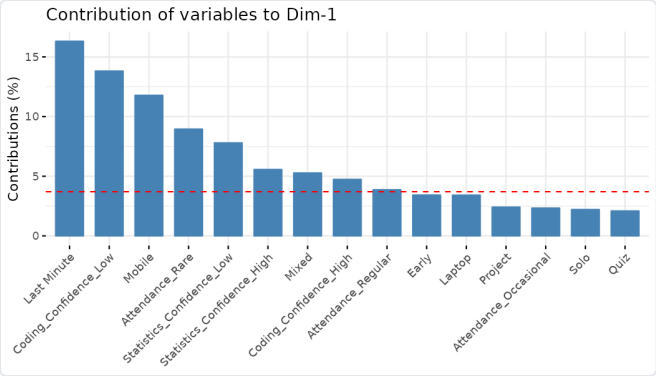


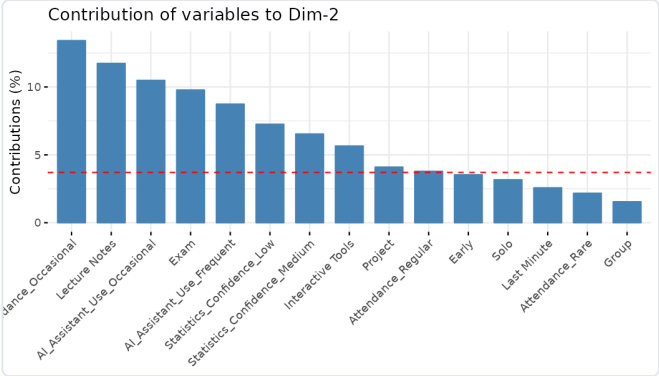
Variable	η² Dim 1	η² Dim 2
Assignment_Timing	**0.544**	0.101
Coding_Confidence	**0.487**	0.026
Primary_Device	**0.400**	0.013
Attendance	**0.397**	**0.311**
Statistics_Confidence	0.361	0.228
Study_Mode	0.229	0.082
Preferred_Assessment	0.132	0.233
Resource_Preference	0.044	**0.297**
AI_Assistant_Use	0.025	**0.313**

4.6 Contributions: who creates the dimension?

```
fviz_contrib(res_mca, choice = "var", axes = 1, top = 15)
fviz_contrib(res_mca, choice = "var", axes = 2, top = 15)
head(res_mca$var$contrib[order(res_mca$var$contrib[,1], decreasing = TRUE), ], 12)
```

- A **large contribution** = the category is important in **constructing** that dimension.
- The **red dashed line** in `fviz_contrib` is the average contribution = 100/27 = 3.7%. Categories above it are "important".
- `order(..., decreasing = TRUE)` sorts the rows by the Dim 1 column, and `head(..., 12)` shows the top 12.





Dimension 1 – top contributions			**Dimension 2 – top contributions**		
Category	**contrib %**	**coord**	**Category**	**contrib %**	**coord**
Last Minute	16.31	+1.53	Attendance_Occasional	13.41	−0.87
Coding_Confidence_Low	13.82	+1.45	Lecture Notes	11.73	+0.76
Mobile	11.78	+1.18	AI_Assistant_Use_Occasional	10.48	+0.67
Attendance_Rare	8.95	+1.88	Exam	9.78	+0.70
Statistics_Confidence_Low	7.79	+1.07	AI_Assistant_Use_Frequent	8.74	−0.51
Statistics_Confidence_High	5.56	−0.57	Statistics_Confidence_Low	7.25	−0.81
Mixed	5.27	−0.59	Statistics_Confidence_Medium	6.54	+0.53
Coding_Confidence_High	4.74	−0.55	Interactive Tools	5.65	−0.55
Attendance_Regular	3.87	−0.40	Project	4.10	−0.37

Checkpoint 3 – answers

Dimension 1

1. Top five: Last Minute, Coding_Confidence_Low, Mobile, Attendance_Rare, Statistics_Confidence_Low.
2. Yes, a clear contrast: the positive side has all "at-risk" categories; the negative side has Regular attendance, Early submission, High confidence, Laptop and Mixed study mode.
3. **Interpretation:** Dim 1 separates **disengaged / at-risk learners** (last-minute, rare attendance, low confidence, mobile only) from **engaged, confident, well-equipped learners**.

Dimension 2

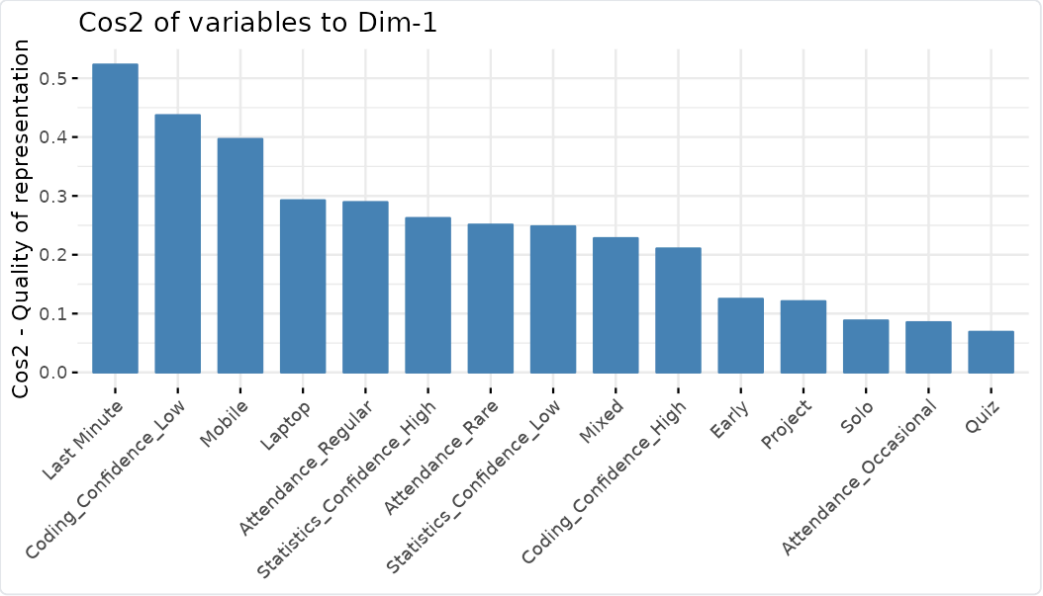
1. Top five: Attendance_Occasional, Lecture Notes, AI_Assistant_Use_Occasional, Exam, AI_Assistant_Use_Frequent.
2. Yes: positive side = Lecture Notes, Exam preference, Occasional AI use, Medium statistics confidence, Solo; negative side = Occasional attendance, Interactive Tools, Frequent AI use, Projects, Low statistics confidence.
3. **Interpretation:** Dim 2 separates a **traditional, exam/notes-based learning style** from an **interactive, project- and AI-assisted style** (often with only occasional attendance).

4.7 cos2: is a category well represented?

```
fviz_cos2(res_mca, choice = "var", axes = 1, top = 15)

category_cos2_2d <- rowSums(res_mca$var$cos2[, 1:2])
sort(category_cos2_2d, decreasing = TRUE)[1:15]
```

- `rowSums(... [,1:2])` adds cos2 on Dim 1 and Dim 2 → quality of representation on the **plane**. Then we sort and show the top 15.



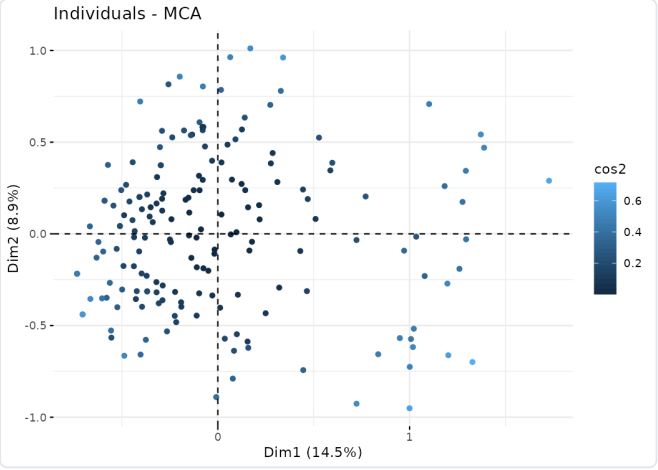
Category	cos2 (Dim 1 + 2)	Category	cos2 (Dim 1 + 2)
Last Minute	0.574	AI_Assistant_Use_Frequent	0.318
Attendance_Regular	0.463	Lecture Notes	0.318
Coding_Confidence_Low	0.440	Laptop	0.305
Mobile	0.407	Attendance_Rare	0.288
Statistics_Confidence_Low	0.390	Statistics_Confidence_High	0.275
Attendance_Occasional	0.386	AI_Assistant_Use_Occasional	0.274

A category may have a **high contribution** but only a **moderate cos2**. Example: Attendance_Rare has the 4th largest contribution to Dim 1 (8.95%), but only 0.288 of it is shown on the plane – because it is rare (12 students) it is far from the origin and also spreads into other dimensions. Interpret it carefully.

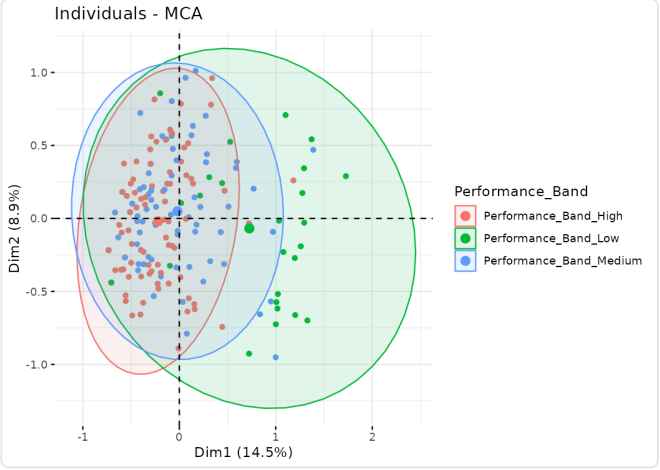
4.8 Individuals (students)

```
fviz_mca_ind(res_mca, geom = "point", col.ind = "cos2")
student_coordinates <- as.data.frame(res_mca$ind$coord)
head(student_coordinates)
```

- `geom = "point"` draws points without labels. **Do not label all 180 students** – the graph becomes unreadable. `col.ind = "cos2"` colours students by how well they are represented.



Students coloured by cos2



Extra: coloured by Performance_Band with `habillage = "Performance_Band", addEllipses = TRUE`

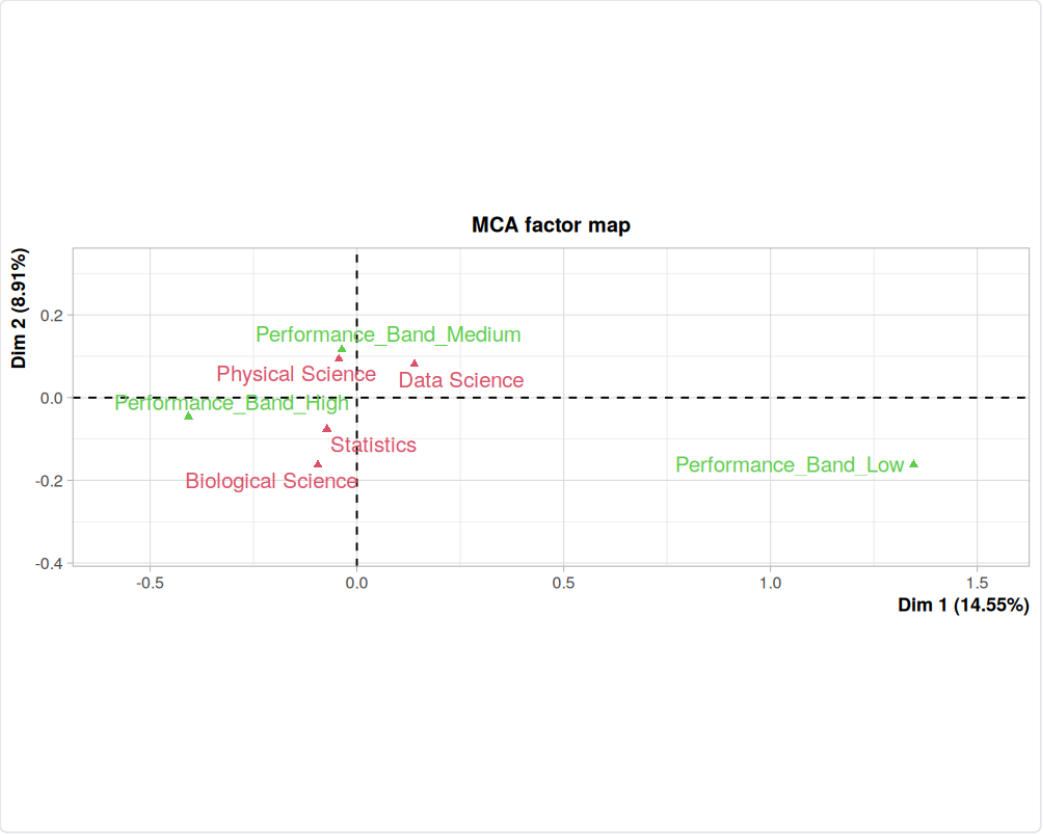
Student	Dim 1	Dim 2	Dim 3	Dim 4	Dim 5
S001	0.157	0.145	0.050	−1.223	−0.091
S002	−0.265	−0.532	−0.391	−0.195	−0.393
S003	−0.369	−0.314	−0.593	−0.315	0.465
S004	−0.705	−0.439	−0.099	−0.005	−0.354
S005	−0.368	0.215	−0.483	−0.161	−0.063
S006	0.587	0.346	0.499	−1.084	−0.346

These numbers are a **reduced numeric representation** of each student's categorical profile – 5 numbers instead of 9 categories. They can be used later for visualisation or clustering.

4.9 Supplementary variables

```
res_mca$quali.sup$coord
plot(res_mca, invisible = c("ind", "var"), hab = "quali")
```

- `invisible = c("ind","var")` hides the students and the active categories, so only the **supplementary categories** are shown. `hab = "quali"` colours them by variable.



Supplementary category	Dim 1 coord	Dim 2 coord	v.test Dim 1	v.test Dim 2
Performance_Band_High	−0.407	−0.046	**−5.04**	−0.56
Performance_Band_Medium	−0.036	0.117	−0.39	1.24
Performance_Band_Low	**1.346**	−0.162	**7.57**	−0.91
Biological Science	−0.094	−0.163	−0.58	−0.99
Data Science	0.139	0.081	1.30	0.75
Physical Science	−0.044	0.093	−0.32	0.69
Statistics	−0.072	−0.076	−0.58	−0.61

η^2 with Dim 1: Performance_Band = 0.349, Programme = 0.010.

Answers to the questions

1. **Performance_Band:** Low is far to the **right** of Dim 1 (+1.35, v.test 7.57), on the same side as last-minute, low-confidence, mobile, rare-attendance categories. High is on the **left** (−0.41, v.test −5.04), with the engaged categories. Medium is near the origin. None is strongly placed on Dim 2.
2. **Programme** is **not** separated: all four programmes are close to the origin, all |v.test| < 2, and $\eta^2 \approx 0.01$. Learning behaviour looks similar across programmes.
3. **Yes**, the behavioural structure found by MCA (Dim 1) is clearly **associated** with performance, even though performance was not used to build the axes.
4. **Not causality:** this is observational data and an association. Other factors (prior knowledge, workload, motivation, health, family) could cause both behaviour and performance, or low performance could cause low confidence (reverse direction). A designed experiment or causal analysis would be needed.

4.10 Describe the dimensions automatically: dimdesc()

```
dimdesc(res_mca, axes = 1:2)
```

For each dimension it gives (1) the **variables** most linked to it (R^2 from a one-way ANOVA of the coordinates on the variable, with p-value), and (2) the **categories** (an "Estimate" – positive = on the positive side – with p-value). Summary for your data:

Dim 1 – variables (R^2)	Dim 2 – variables (R^2)
Assignment_Timing 0.544, Coding_Confidence 0.487, Primary_Device 0.400, Attendance 0.397, Statistics_Confidence 0.361, **Performance_Band 0.349**, Study_Mode 0.229, Preferred_Assessment 0.132, Resource_Preference 0.044	AI_Assistant_Use 0.313, Attendance 0.311, Resource_Preference 0.297, Preferred_Assessment 0.233, Statistics_Confidence 0.228, Assignment_Timing 0.101, Study_Mode 0.082
+ Last Minute, Coding Low, Mobile, Performance Low, Attendance Rare, Stats Low, Solo, Quiz … − Laptop, Attendance Regular, Stats High, Mixed, Coding High, Performance High, Early, Project	+ Lecture Notes, AI Occasional, Exam, Attendance Regular, Stats Medium, Early, Solo … − AI Frequent, Attendance Occasional, Stats Low, Interactive Tools, Project, Last Minute, Group, Videos

Use `dimdesc` as a **diagnostic aid**, not as a substitute for your own interpretation.

4.11 Optional comparison: Indicator vs Burt approach

```
res_indicator <- MCA(dat, quali.sup = sup_idx, method = "Indicator", graph = FALSE)
res_burt      <- MCA(dat, quali.sup = sup_idx, method = "Burt",      graph = FALSE)
res_indicator$eig[1:5, ]
res_burt$eig[1:5, ]
```

Dim	Indicator eigenvalue	Indicator %	Burt eigenvalue	Burt %
1	0.2910	14.55	0.0847	30.14
2	0.1783	8.91	0.0318	11.31
3	0.1557	7.78	0.0242	8.63
4	0.1481	7.41	0.0219	7.81
5	0.1361	6.81	0.0185	6.60

- The **Indicator** method (default) analyses the 180 × 27 0/1 table. The **Burt** method analyses the 27 × 27 Burt table (all two-way cross-tables of categories).
- **What changes:** Burt eigenvalues are the **squares** of the indicator eigenvalues ($0.2910^2 = 0.0847$), so the percentages look much bigger (30% vs 14.5%) and the scaling of category coordinates changes.
- **What stays the same:** the **directions** of the dimensions, the order of dimensions and the relative positions of categories – the interpretation is substantively similar.
- Check `?MCA` before drawing conclusions about eigenvalue scaling.

4.12 Mini-report – a model answer (about 300 words)

Why MCA? All nine active variables (study mode, resources, attendance, assignment timing, confidence levels, AI use, assessment preference and device) are categorical. PCA needs quantitative variables and Euclidean distances; coding categories as numbers would invent distances. MCA analyses the indicator table of categories directly.

How many dimensions? Dim 1 explains 14.5% and Dim 2 8.9% of inertia (23.5% together). The scree plot drops clearly after Dim 1 and then declines slowly. I keep **two dimensions for interpretation**, but would keep about five for clustering, since MCA percentages are naturally low.

Dimension 1 – "Engagement / at-risk learning behaviour". The positive side is formed by Last Minute submission (16.3%), Low coding confidence (13.8%), Mobile device (11.8%), Rare attendance (9.0%) and Low statistics confidence (7.8%); the negative side by Regular attendance, Early submission, High confidence, Laptop and Mixed study.

Dimension 2 – "Traditional vs interactive learning style". Lecture Notes, Exam preference and Occasional AI use are on the positive side; Occasional attendance, Interactive Tools, Frequent AI use and Projects on the negative side.

High contributions: Last Minute, Coding_Confidence_Low and Mobile (Dim 1); Attendance_Occasional and Lecture Notes (Dim 2).

Representation: the key categories are moderately well represented on the plane: Last Minute cos2 = 0.57, Coding_Confidence_Low 0.44, Mobile 0.41, Attendance_Occasional 0.39. Attendance_Rare (0.29) is rarer and less well represented, so it is interpreted with care.

Performance_Band (supplementary): Low performance lies far on the positive side of Dim 1 (v.test 7.6) and High on the negative side (v.test −5.0); Programme is near the origin. So the behavioural pattern is associated with performance, but this does not prove that behaviour causes performance.

Include: the scree plot (4.4), the category map (4.5) and a contribution plot (4.6).

5. FAMD – the idea

Factor Analysis of Mixed Data (FAMD) is used when the data have **both** quantitative and categorical variables. It is a combination of PCA and MCA:

- **Quantitative variables** are **standardised** (like PCA on the correlation matrix).
- **Categorical variables** are coded as indicator columns and **weighted** as in MCA.
- This **balances** the two types so that neither dominates just because of its coding.
- Each dimension is linked to quantitative variables through the **squared correlation** and to categorical variables through **η² (correlation ratio)**. The eigenvalue of a dimension = the sum of these values.

6. FAMD step by step

Data: `graduate_employability_mixed.csv` – 180 graduates. (The two uploaded copies, "graduate_employability_mixed.csv" and "... 1.csv", are identical.)

6.1 Read and prepare

```
dat <- read.csv("graduate_employability_mixed.csv", stringsAsFactors = TRUE)
str(dat)
head(dat)
rownames(dat) <- dat$Graduate_ID
dat$Graduate_ID <- NULL
head(dat)
sapply(dat, class)
```

6.2 Part C Task 1 – separate quantitative and categorical variables

```
num_vars <- names(dat)[sapply(dat, is.numeric)]
num_vars;  length(num_vars)     # 8
cat_vars <- names(dat)[sapply(dat, is.factor)]
cat_vars;  length(cat_vars)     # 6
```

- `sapply(dat, is.numeric)` gives TRUE for numeric/integer columns; `names(dat)[...]` keeps the names where TRUE. This works here because the categorical columns are real **factors** (text), unlike vs/am in mtcars last week.

Quantitative (8)	Categorical (6)
GPA, Programming_Score, Statistics_Score, Communication_Score, Internship_Months, Portfolio_Projects, Weekly_SelfStudy_Hours, Hackathons	Degree_Stream (4 levels), Internship_Sector (5), English_Level (3), Teamwork_Level (3), Career_Preference (5), **Employment_Status** (4) – supplementary

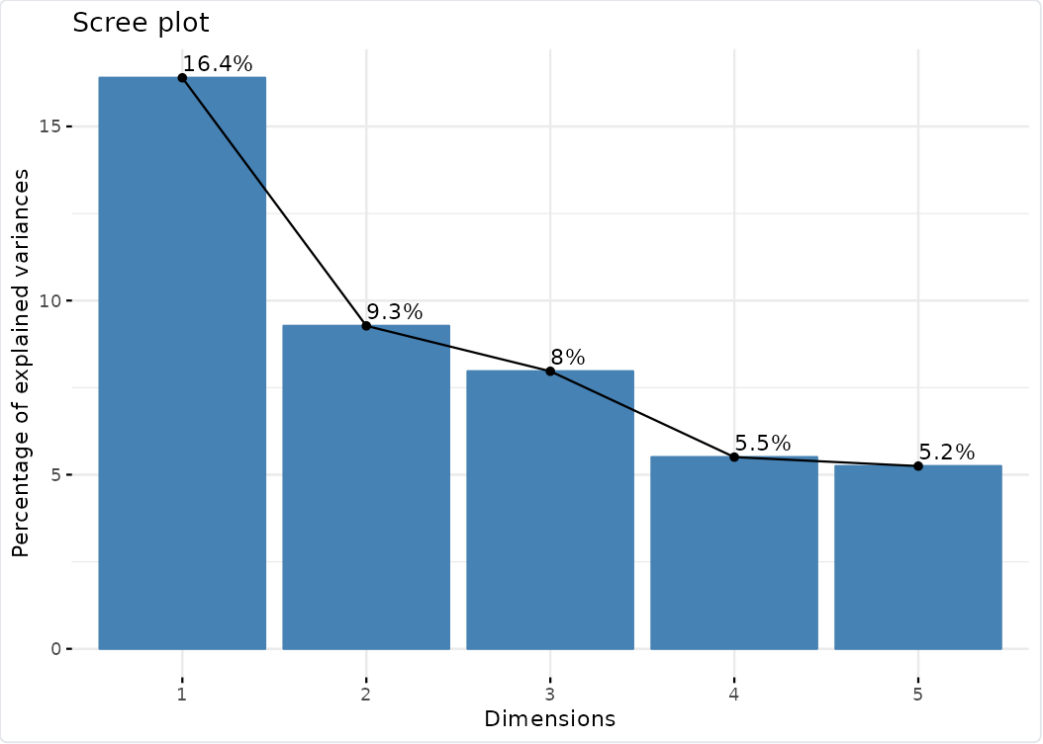
Frequencies: Degree_Stream – Bioinformatics 11, Computer Science 50, Data Science 62, Statistics 57. English_Level – Advanced 115, Basic 20, Intermediate 45. Teamwork – Competent 69, Developing 23, Strong 88. Employment_Status – Employed-Data Role 103, Employed-Other 26, Higher Studies 23, Seeking 28. (Means: GPA 3.31, Programming 72.4, Statistics 75.7, Communication 75.2.)

6.3 Run FAMD (Task 2)

```
sup_idx <- which(names(dat) == "Employment_Status")     # 14
res_famd <- FAMD(dat, sup.var = sup_idx, ncp = 5, graph = FALSE)
res_famd$eig
fviz_eig(res_famd, addlabels = TRUE)
```

- In `FAMD()` the argument is `sup.var` (it can be quantitative or categorical). Employment_Status is the **outcome**, so we keep it supplementary: it does not build the axes, we only see where it falls.

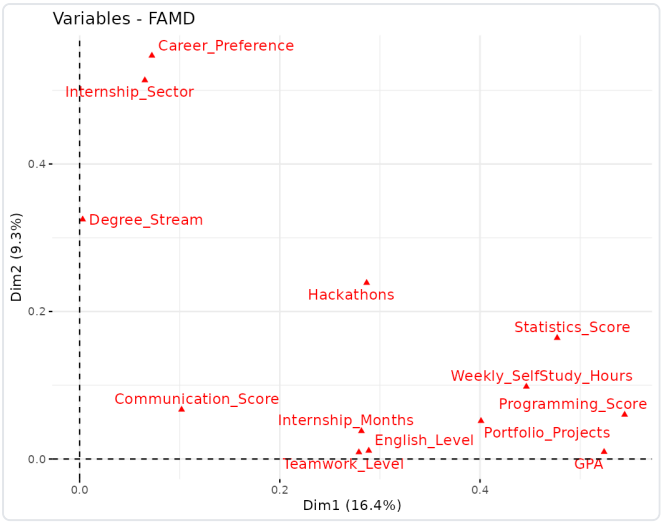
Dim	Eigenvalue	%	Cumulative %
1	3.771	16.40	16.40
2	2.133	9.27	25.67
3	1.833	7.97	33.64
4	1.266	5.50	39.14
5	1.207	5.25	44.39



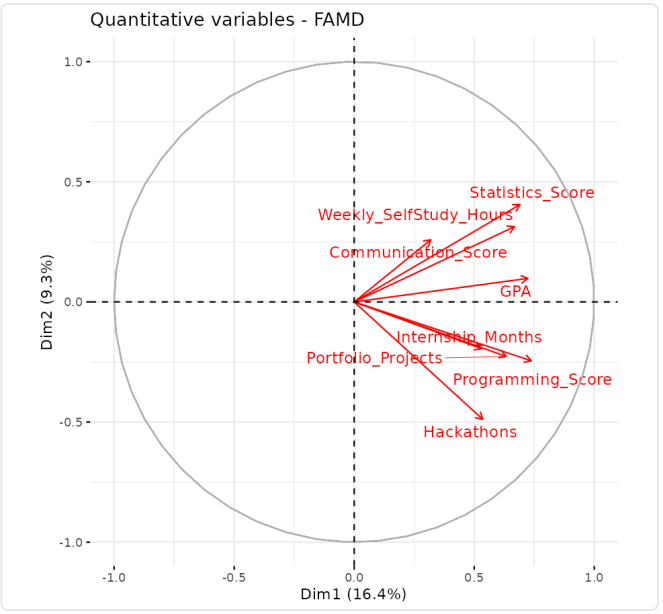
Dim 1 is clearly the strongest; after Dim 3 the values flatten (elbow around 3). Two dimensions explain 25.7%, three 33.6%.

6.4 Variables

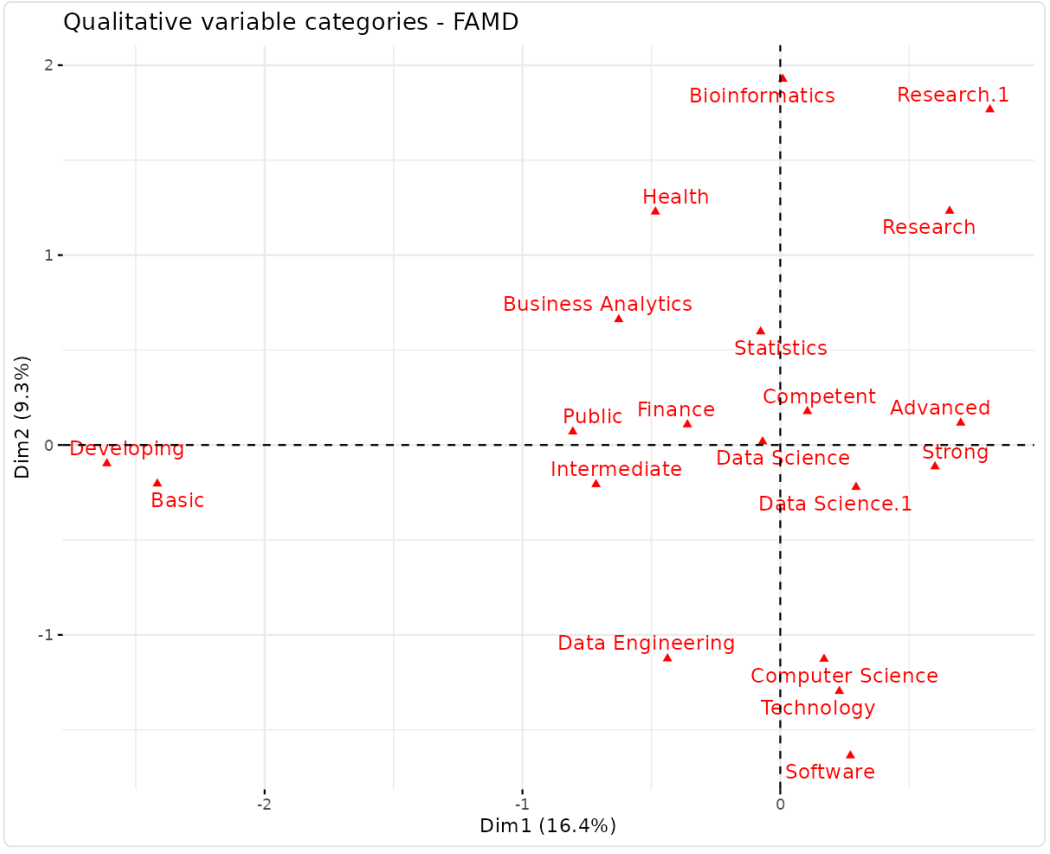
```
fviz_famd_var(res_famd, repel = TRUE)                          # all variables
fviz_famd_var(res_famd, choice = "quanti.var", repel = TRUE)   # correlation circle
fviz_famd_var(res_famd, choice = "quali.var",  repel = TRUE)   # categories
```



All variables: coordinates = squared correlation (numeric) or η² (categorical)



Correlation circle of quantitative variables



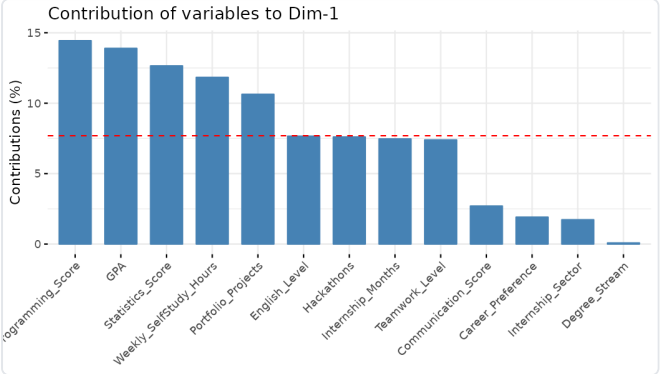
Categories of the qualitative variables

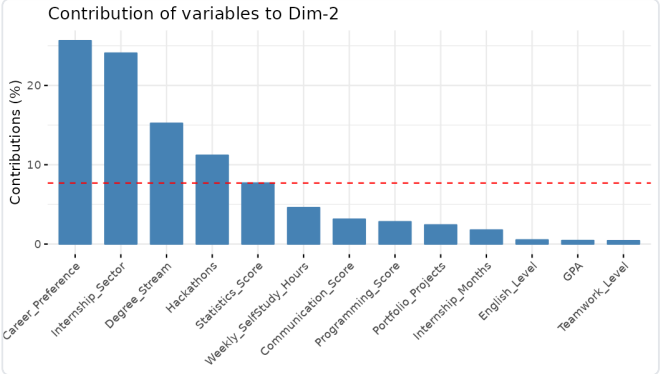
Tip: duplicate category names

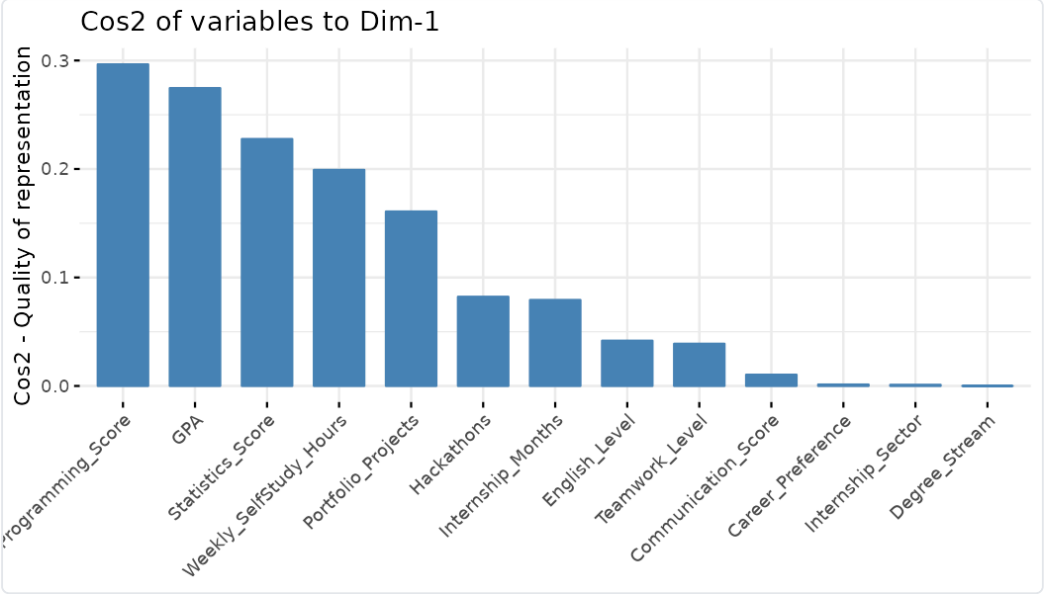
Degree_Stream and Career_Preference both have a "Data Science" category, and Internship_Sector and Career_Preference both have "Research". FactoMineR renames the second ones to **"Data Science.1"** and **"Research.1"** (these are Career_Preference). To avoid confusion you can rename levels first, e.g. `levels(dat$Career_Preference) <- paste0("Career_", levels(dat$Career_Preference))`.

6.5 Contributions and representation (Tasks 3, 4, 5)

```
fviz_contrib(res_famd, choice = "var", axes = 1, top = 15)
fviz_cos2(res_famd,    choice = "var", axes = 1, top = 15)
fviz_contrib(res_famd, choice = "var", axes = 2, top = 15)
```







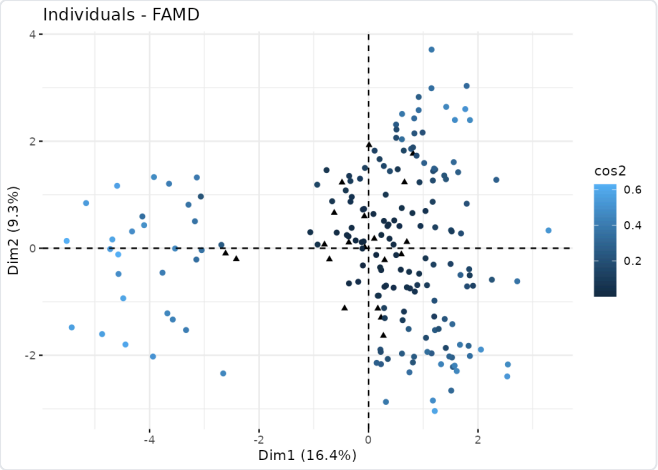
Variable	Contrib Dim 1 %	Contrib Dim 2 %	Correlation with Dim 1	Correlation with Dim 2
Programming_Score	**14.44**	2.82	0.738	−0.245
GPA	**13.89**	0.45	0.724	0.098
Statistics_Score	**12.65**	7.70	0.691	0.405
Weekly_SelfStudy_Hours	11.83	4.60	0.668	0.313
Portfolio_Projects	10.63	2.42	0.633	−0.227
English_Level	7.66	0.53	η² 0.289	η² 0.011
Hackathons	7.60	11.20	0.535	−0.489
Internship_Months	7.47	1.77	0.531	−0.195
Teamwork_Level	7.40	0.43	η² 0.279	η² 0.009
Communication_Score	2.70	3.14	0.319	0.259
Career_Preference	1.91	**25.64**	η² 0.072	η² 0.547
Internship_Sector	1.73	**24.08**	η² 0.065	η² 0.514
Degree_Stream	0.08	**15.23**	η² 0.003	η² 0.325

Answers

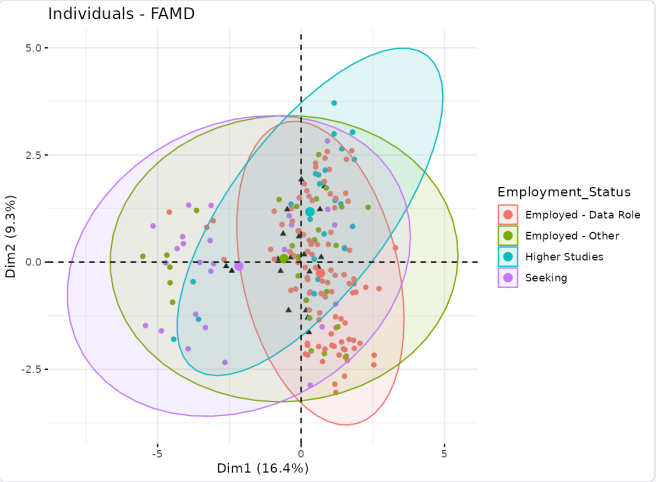
- **Task 3 – three most important quantitative variables for Dim 1:** Programming_Score (14.4%), GPA (13.9%), Statistics_Score (12.7%). All are positively correlated with Dim 1 (≈ 0.7).
- **Task 4 – most important qualitative categories for Dim 1:** Teamwork_Level = **Developing** (contrib 6.1%, coord −2.61) and English_Level = **Basic** (4.6%, −2.42) on the negative side; English_Level = **Advanced** (2.2%, +0.70) and Teamwork = **Strong** (1.2%, +0.60) on the positive side. As whole variables, English_Level (7.7%) and Teamwork_Level (7.4%).
- So **Dim 1 = "overall academic and employability strength"**: high GPA, strong programming/statistics scores, more self-study, projects, hackathons and internships, advanced English and strong teamwork (right) vs weaker profiles with basic English and developing teamwork (left).
- **Task 5 – Dim 2 = "research/health orientation vs technology/software orientation"**. Positive: Career = Research, Internship sector = Research / Health, Degree = Bioinformatics / Statistics, higher Statistics score and self-study hours. Negative: Internship sector = Technology, Career = Software / Data Engineering, Degree = Computer Science, more Hackathons. It is mostly built by the **categorical** variables (Career 25.6%, Sector 24.1%, Degree 15.2%).

6.6 Individuals

```
fviz_famd_ind(res_famd, geom = "point", col.ind = "cos2")
```



Graduates coloured by cos2



Extra: coloured by Employment_Status (`habillage = "Employment_Status", addEllipses = TRUE`)

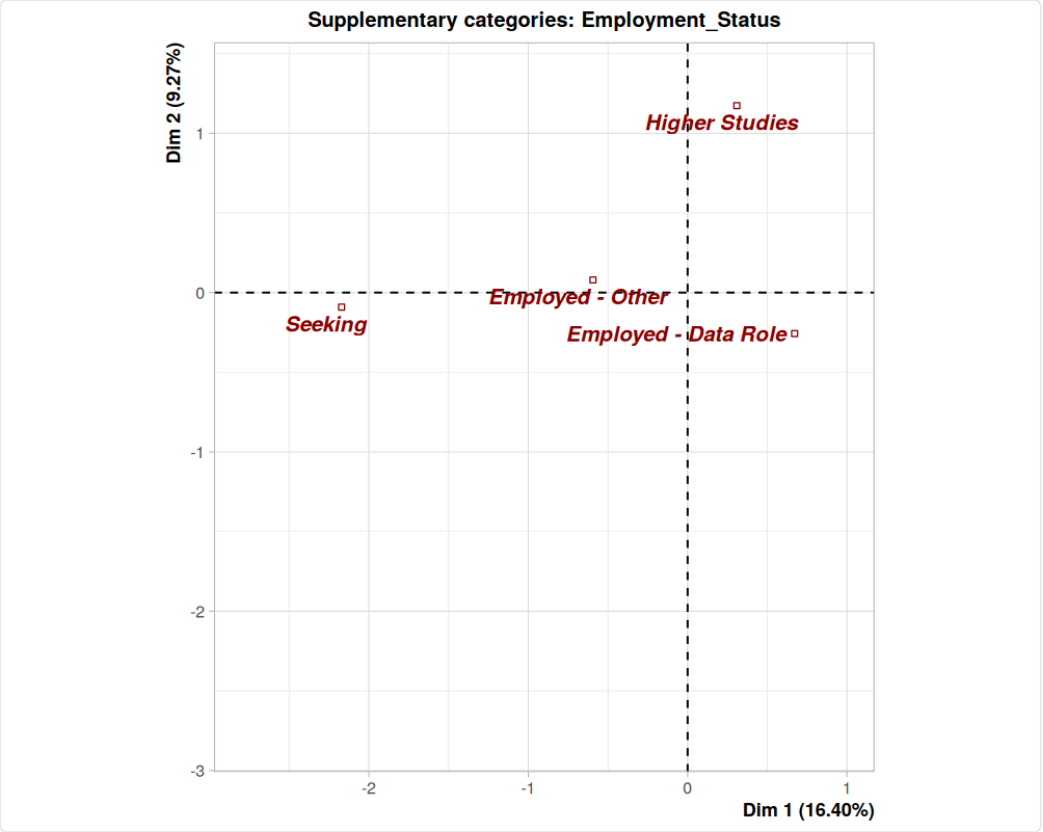
6.7 Supplementary outcome (Task 6)

```
# from the lecture code — gives an ERROR in current factoextra:
fviz_famd_var(res_famd, choice = "quali.sup", repel = TRUE)
# Error: 'arg' should be one of "var", "quanti.var", "quali.var"

# working alternative:
plot(res_famd, choix = "ind", invisible = c("ind", "quali"),
     title = "Supplementary categories: Employment_Status")
res_famd$quali.sup$coord
res_famd$quali.sup$v.test
```

Code fix

`fviz_famd_var()` only accepts `"var"`, `"quanti.var"` or `"quali.var"`. Use the base `plot()` above (hides individuals and active categories, shows only supplementary ones), or colour individuals with `habillage` as in 6.6.



Employment_Status	Dim 1	Dim 2	v.test Dim 1	v.test Dim 2
Employed – Data Role	0.671	−0.257	**5.35**	−2.73
Employed – Other	−0.594	0.079	−1.68	0.30
Higher Studies	0.309	**1.173**	0.81	**4.11**
Seeking	**−2.171**	−0.091	**−6.42**	−0.36

Interpretation: "Seeking" graduates are far on the **weak** side of Dim 1; "Employed – Data Role" on the **strong** side. "Higher Studies" is high on Dim 2 (research/health orientation). η^2 of Employment_Status: Dim 1 = 0.28, Dim 2 = 0.10. Again: association, not causation.

6.8 dimdesc for FAMD

```
dimdesc(res_famd, axes = 1:2)
```

- **Dim 1:** correlations – Programming 0.74, GPA 0.72, Statistics 0.69, Self-study 0.67, Portfolio 0.63, Hackathons 0.54, Internship months 0.53, Communication 0.32. Categorical – English_Level (R^2 0.29), Employment_Status (0.28), Teamwork (0.28). Positive categories: English Advanced, Employed-Data Role, Teamwork Strong; negative: Teamwork Developing, Seeking, English Basic.
- **Dim 2:** correlations – Statistics 0.41, Self-study 0.31, Communication 0.26; negative: Hackathons −0.49, Programming −0.25, Portfolio −0.23. Categorical – Career_Preference (R^2 0.55), Internship_Sector (0.51), Degree_Stream (0.32). Positive: Career Research, Sector Research, Bioinformatics, Higher Studies, Sector Health; negative: Sector Technology, Career Software, Computer Science, Career Data Engineering.

7. Using the reduced coordinates for clustering (Task 7, 8)

```
graduate_coordinates <- as.data.frame(res_famd$ind$coord)
graduate_coordinates             # 180 rows x 5 columns (Dim.1 ... Dim.5)
X <- graduate_coordinates[, 1:3] # first three coordinates
```

Graduate	Dim.1	Dim.2	Dim.3
G001	1.526	−2.042	0.911
G002	0.988	2.161	0.867
G003	1.909	−0.702	1.530
G004	−0.235	0.144	−1.359
G005	0.811	−2.134	1.315
G006	−4.100	0.430	0.937

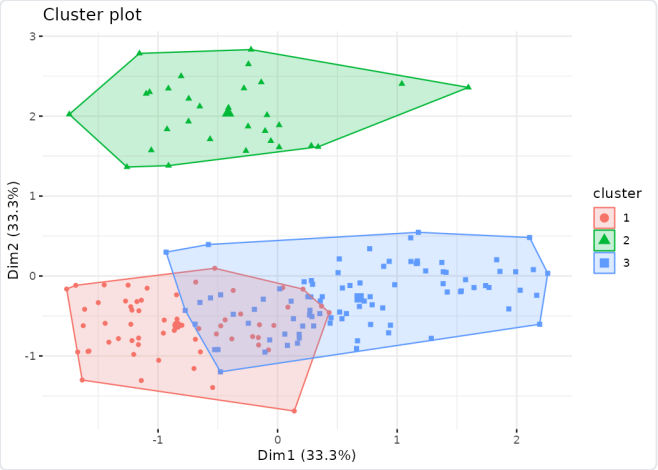
Task 7: `graduate_coordinates[, 1:3]` gives each graduate's coordinates on Dims 1–3 (the first rows are shown above).

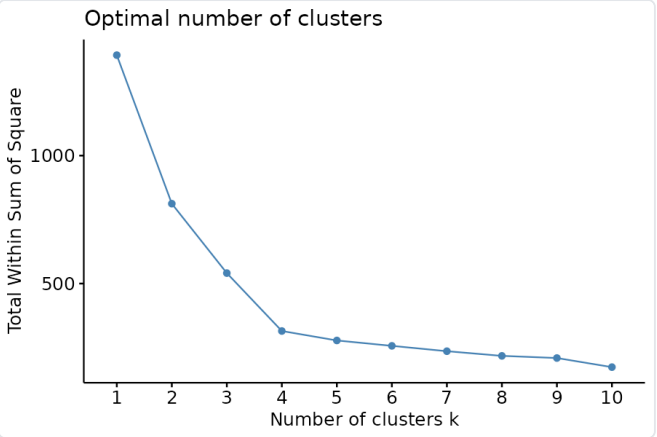
Task 8 – how they can be used for clustering

- The 13 mixed variables (numbers + categories) are replaced by **3 numeric, uncorrelated, comparable** coordinates. Ordinary distance-based methods (**k-means**, **hierarchical clustering**) can use them directly with Euclidean distance – something they cannot do on raw categories.
- The first dimensions keep the main structure and drop much of the **noise** in later dimensions.
- After clustering, we **describe each cluster** using the original variables and the dimension meanings (e.g. "strong academic, technology-oriented").

Demonstration (k-means with 3 clusters on Dims 1–3, `set.seed(123)`, `nstart = 25`):

```
set.seed(123)
km <- kmeans(X, centers = 3, nstart = 25)
km$size                                  # 59 30 91
table(km$cluster, dat$Employment_Status)
fviz_cluster(km, data = X, geom = "point")
```





Elbow plot for choosing k

Cluster (size)	Centre (Dim1, Dim2, Dim3)	Data Role	Other	Higher Studies	Seeking	Meaning
1 (59)	1.20, −1.48, 0.46	47	6	2	4	Strong, technology/software-oriented
2 (30)	−3.97, −0.19, 0.56	4	6	3	**17**	Weaker academic/employability profile
3 (91)	0.53, 1.02, −0.48	52	14	18	7	Moderate-strong, research/analytics-oriented

Employment_Status was **not** used to make the clusters, yet cluster 2 contains most "Seeking" graduates.

8. Self-Learning Activity – answers

Part A – Method selection

Situation	Method	Justification
1. 12 continuous laboratory measurements	**PCA**	All variables are quantitative, so PCA (usually on standardised data) is the natural choice.
2. Survey with 18 Yes/No and Likert questions	**MCA**	Several categorical variables; MCA analyses them without inventing numeric distances.
3. Contingency table of district × transport mode	**CA**	A single two-way table of counts is exactly what Correspondence Analysis is designed for.
4. GPA, test scores, degree stream, sector, career preference	**FAMD**	A mix of quantitative and categorical variables; FAMD balances both types.
5. Genetics, clinical, imaging and lifestyle blocks	**MFA**	Variables come in meaningful groups; MFA balances the blocks so no single block dominates.

Part B – Categorical data set

1. **Frequency summary:** `lapply(dat, table)` – see the table in 4.2.
2. **First five eigenvalues:** 0.2910, 0.1783, 0.1557, 0.1481, 0.1361 (14.55%, 8.91%, 7.78%, 7.41%, 6.81%).
3. **Scree plot:** `fviz_eig(res_mca, addlabels = TRUE)` – see 4.4.
4. **Top 5 for Dim 1:** Last Minute (16.3%), Coding_Confidence_Low (13.8%), Mobile (11.8%), Attendance_Rare (9.0%), Statistics_Confidence_Low (7.8%).
5. **Top 5 for Dim 2:** Attendance_Occasional (13.4%), Lecture Notes (11.7%), AI_Assistant_Use_Occasional (10.5%), Exam (9.8%), AI_Assistant_Use_Frequent (8.7%).
6. **High quality on the first two dimensions (cos2):** Last Minute 0.57, Attendance_Regular 0.46, Coding_Confidence_Low 0.44, Mobile 0.41, Statistics_Confidence_Low 0.39.
7. **Names:** the lecture code uses "Dim 1 – Learning Behaviour and Study Habits" and "Dim 2 – Learning Preferences and Engagement". More specific names from the output: **Dim 1 = Engaged vs at-risk learning behaviour**; **Dim 2 = Traditional (notes/exam) vs interactive (tools/projects/AI) learning style**.
8. **Performance_Band:** Low lies on the at-risk side of Dim 1 (coord 1.35, v.test 7.6), High on the engaged side (−0.41, v.test −5.0), Medium near the centre. This shows an **association** between behaviour patterns and performance; it does not show that the behaviour **causes** the performance.

Part C – Mixed data set

1. **Lists:** 8 quantitative (GPA, Programming_Score, Statistics_Score, Communication_Score, Internship_Months, Portfolio_Projects, Weekly_SelfStudy_Hours, Hackathons); 6 categorical (Degree_Stream, Internship_Sector, English_Level, Teamwork_Level, Career_Preference, Employment_Status).

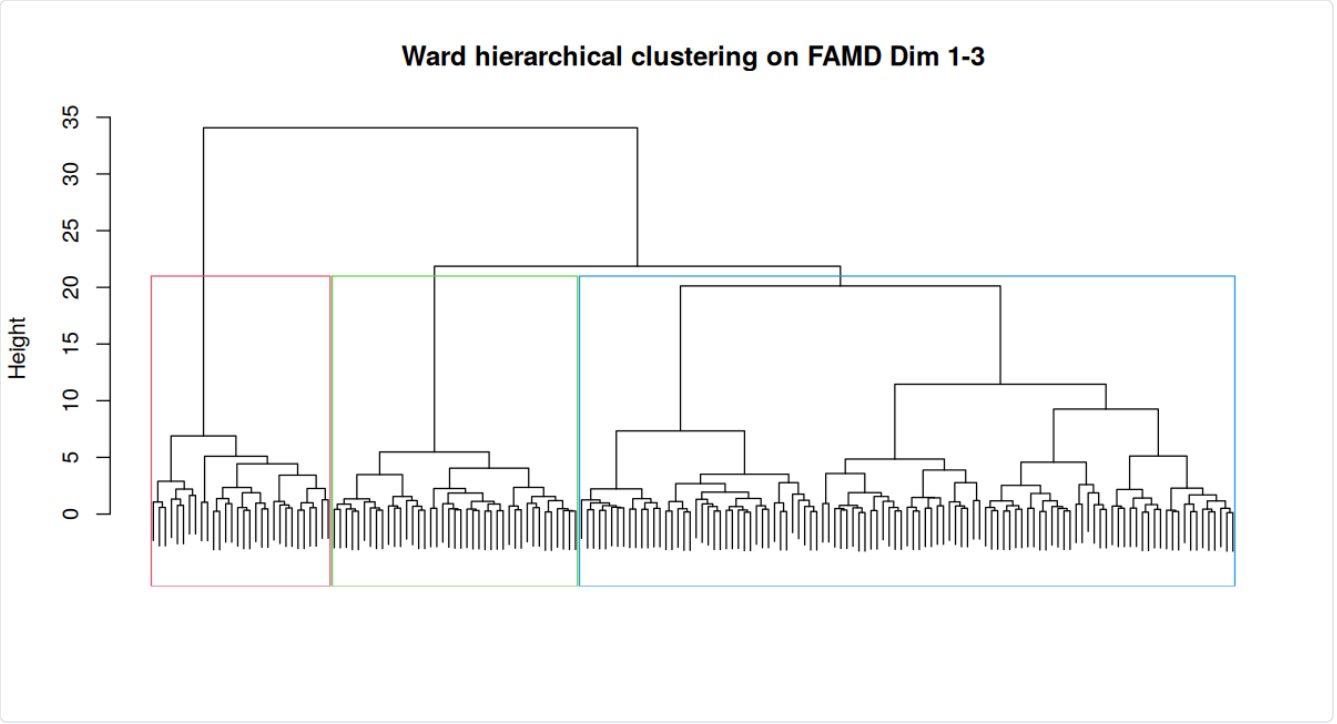
2. **FAMD + scree plot:** see 6.3 (Dim 1 16.4%, Dim 2 9.3%, Dim 3 8.0%).
3. **Top quantitative for Dim 1:** Programming_Score, GPA, Statistics_Score.
4. **Top qualitative categories for Dim 1:** Teamwork = Developing and English = Basic (negative), English = Advanced and Teamwork = Strong (positive).
5. **Dim 2:** research/health/bioinformatics orientation vs technology/software/computer-science orientation (see 6.5).
6. **Supplementary employment categories:** see 6.7 – Seeking far negative on Dim 1, Data Role positive on Dim 1, Higher Studies high on Dim 2.
7. **First three coordinates:** `graduate_coordinates[, 1:3]` (see Section 7).
8. **Use in clustering:** see Section 7.

Part D – Critical thinking

1. **Contribution vs cos2:** contribution tells how much a category **builds** an axis (importance for the axis); cos2 tells how **well** that category is **shown** by the axis (quality). A rare category can contribute a lot yet be only partly represented, and a common category near the origin can be well represented but contribute little.
2. **Far from the origin:** it has a profile very different from the average student, so it often helps define a dimension and points to a distinctive group (e.g. Last Minute, Attendance_Rare).
3. **Rare categories:** they are pushed far from the origin and can get large contributions based on only a few people. Their position is unstable (a few students can change it), so interpret carefully; consider merging very rare categories.
4. **Two dimensions are not "the data":** here they keep only 23.5% (MCA) or 25.7% (FAMD) of the inertia. Much structure lies in later dimensions; the 2-D map is a simplified view.
5. **When dimension reduction harms prediction:** e.g. predicting employment where an important predictor (say Hackathons or a rare category) has little variance and falls into a later dimension that we drop. PCA/MCA/FAMD choose axes by **variance**, not by relation to the outcome, so the information needed for prediction can be thrown away.

Optional challenge – why clustering results depend on…

- **Number of retained dimensions:** 2 vs 3 vs 5 dimensions give different distances between people (more dims add detail but also noise), so clusters change.
- **Scaling / weighting:** distances depend on the scale of each coordinate; Dim 1 has the largest spread so it dominates. Changing weights (e.g. standardising coordinates, or the categorical/quantitative balance in FAMD) changes the clusters.
- **Chosen number of clusters:** k-means always gives exactly k groups; use an elbow plot (within-SS: 1393, 812, 541, 315, … for k = 1–4) or silhouette, and subject knowledge.
- **Quality of representation:** individuals with low cos2 are poorly shown in the retained dimensions, so their cluster membership is less reliable.



Comparison: Ward hierarchical clustering on Dims 1–3 (cut into 3 groups: 41, 109, 30)

9. Introduction to cluster analysis

Cluster analysis is unsupervised learning: it puts similar observations into groups (clusters) using a **distance** measure. The lecture started with a tiny example.

9.1 The data

```
names <- list(c("g1","g2","g3","g4","g5"), c("p1","p2"))
sl.clus.dat <- matrix(c(1,1, 1,1.1, 3,2, 3,2.3, 5,5), ncol = 2,
                      byrow = TRUE, dimnames = names)
plot(sl.clus.dat)
plot(sl.clus.dat, type = "n", xlim = c(0,6), ylim = c(0,6))
text(sl.clus.dat, labels = row.names(sl.clus.dat))
```

- 5 points (g1 … g5) with 2 variables (p1, p2). `byrow = TRUE` fills the matrix row by row; `dimnames` gives row and column names.
- `type = "n"` draws an empty plot (no points), then `text()` writes each point's name at its position.

	p1	p2
g1	1	1.0
g2	1	1.1
g3	3	2.0
g4	3	2.3
g5	5	5.0

9.2 Distance matrix

```
print(dist(sl.clus.dat, method = "euclidean"), digits = 3)
```

Euclidean distance = $\sqrt{((p1_i - p1_j)^2 + (p2_i - p2_j)^2)}$. Example g1–g3: $\sqrt{(2^2 + 1^2)} = \sqrt{5} = 2.24$.

	g1	g2	g3	g4
g2	0.10			
g3	2.24	2.19		
g4	2.39	2.33	0.30	
g5	5.66	5.59	3.61	3.36

9.3 Hierarchical clustering – single and complete linkage

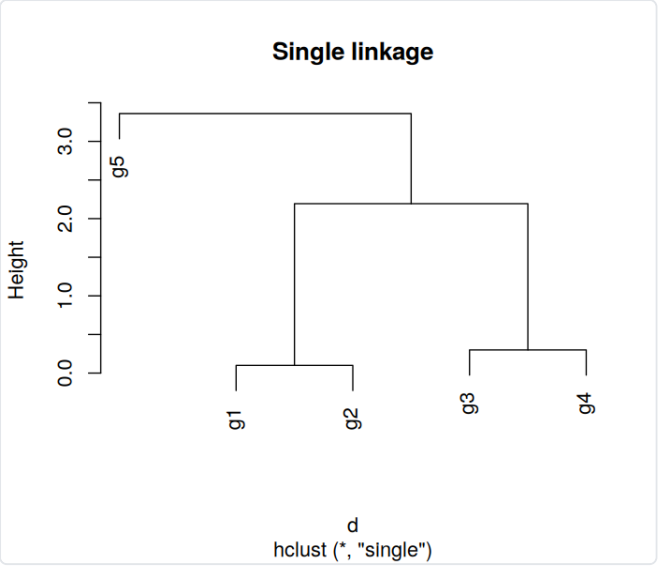
```
d = dist(sl.clus.dat, method = "euclidean")
hclust_out = hclust(d, method = "single")
hclust_out
plot(hclust_out)

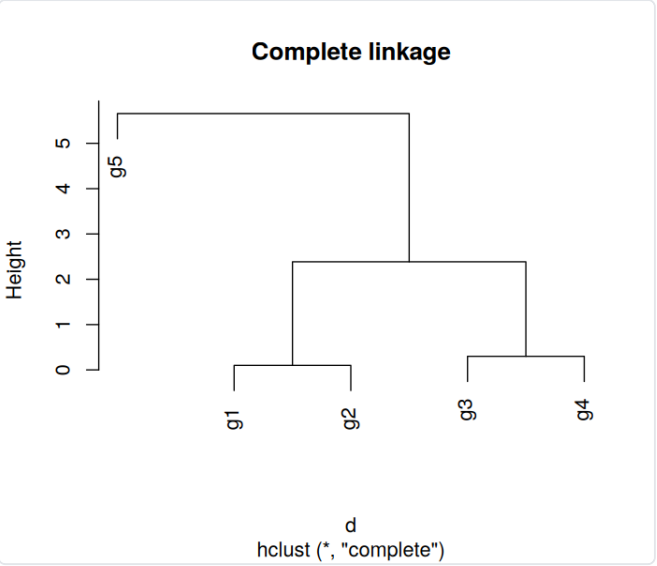
hclust_out_complete = hclust(d, method = "complete")
plot(hclust_out_complete)
```

Agglomerative hierarchical clustering: start with every point as its own cluster, then repeatedly **join the two closest clusters** until one cluster remains. The **linkage** decides the distance between two clusters:

- **Single linkage** = the **minimum** distance between any two members (nearest neighbours).
- **Complete linkage** = the **maximum** distance between any two members (furthest neighbours).

Step	Single linkage	Complete linkage
1	Join g1, g2 at **0.10**	Join g1, g2 at **0.10**
2	Join g3, g4 at **0.30**	Join g3, g4 at **0.30**
3	{g1,g2}–{g3,g4} = min(2.24, 2.39, 2.19, 2.33) = **2.19** → join	{g1,g2}–{g3,g4} = max(2.24, 2.39, 2.19, 2.33) = **2.39** → join
4	g5 to the rest = min(5.66, 5.59, 3.61, 3.36) = **3.36** → join	g5 to the rest = max(5.66, 5.59, 3.61, 3.36) = **5.66** → join





The **dendrogram** (tree) shows the order of joining; the height of each join is the distance at which the clusters were merged. Here both linkages give the same order, but the heights of the last two joins differ (2.19/3.36 vs 2.39/5.66). Cutting the tree at a chosen height gives the clusters, e.g. {g1,g2}, {g3,g4}, {g5}.

Next week
Run different distances and linkages (e.g. Manhattan, average, Ward) on a new example.

10. Summary and full R workflow

Key points

- PCA → quantitative; CA → one contingency table; **MCA → several categorical**; **FAMD → mixed**; MFA → blocks.
- Prepare data: IDs to row names, all categorical columns as **factors**, look at frequencies (watch rare categories).
- Supplementary variables are projected but do **not** build the axes – good for checking outcomes like performance or employment.
- Read: **eigenvalues/scree** → how many dims; **contributions** → what builds an axis; **cos2** → how well it is shown; **v.test/dimdesc** → significance and automatic description.
- MCA percentages are low by nature – don't use an automatic 80% rule.
- Give each axis a meaningful **name**; association ≠ causation.
- Reduced coordinates (`$ind$coord`) can feed **clustering**; results depend on dims kept, scaling, k and representation quality.
- Hierarchical clustering: distance matrix → linkage (single = min, complete = max) → dendrogram.

```r
## ---------- MCA ----------
dat <- read.csv("student_learning_profiles_categorical.csv", stringsAsFactors = TRUE)
rownames(dat) <- dat$Student_ID;  dat$Student_ID <- NULL
dat[] <- lapply(dat, as.factor)
library(FactoMineR); library(factoextra)
sup_idx <- which(names(dat) %in% c("Programme", "Performance_Band"))
res_mca <- MCA(dat, quali.sup = sup_idx, ncp = 5, graph = FALSE)
res_mca$eig;  fviz_eig(res_mca, addlabels = TRUE)
fviz_mca_var(res_mca, choice = "var.cat", repel = TRUE)
fviz_contrib(res_mca, choice = "var", axes = 1, top = 15)
fviz_cos2(res_mca, choice = "var", axes = 1, top = 15)
fviz_mca_ind(res_mca, geom = "point", col.ind = "cos2")
res_mca$quali.sup$coord;  plot(res_mca, invisible = c("ind", "var"), hab = "quali")
dimdesc(res_mca, axes = 1:2)
student_coordinates <- as.data.frame(res_mca$ind$coord)

## ---------- FAMD ----------
dat <- read.csv("graduate_employability_mixed.csv", stringsAsFactors = TRUE)
rownames(dat) <- dat$Graduate_ID;  dat$Graduate_ID <- NULL
sup_idx <- which(names(dat) == "Employment_Status")
res_famd <- FAMD(dat, sup.var = sup_idx, ncp = 5, graph = FALSE)
res_famd$eig;  fviz_eig(res_famd, addlabels = TRUE)
fviz_famd_var(res_famd, choice = "quanti.var", repel = TRUE)
fviz_famd_var(res_famd, choice = "quali.var",  repel = TRUE)
fviz_contrib(res_famd, choice = "var", axes = 1, top = 15)
fviz_famd_ind(res_famd, geom = "point", col.ind = "cos2")
plot(res_famd, choix = "ind", invisible = c("ind", "quali"))   # supplementary
dimdesc(res_famd, axes = 1:2)
graduate_coordinates <- as.data.frame(res_famd$ind$coord)

## ---------- Hierarchical clustering ----------
d <- dist(sl.clus.dat, method = "euclidean")
plot(hclust(d, method = "single"));  plot(hclust(d, method = "complete"))
```

Further reading: FactoMineR reference manual – cran.r-project.org/web/packages/FactoMineR/FactoMineR.pdf; factoextra reference manual – cran.r-project.org/web/packages/factoextra/factoextra.pdf.